

Machine Learning & Dataset Technical Documentation

EXECUTIVE SUMMARY & DEPLOYED MODEL ACCURACY

Primary Algorithm: Polynomial Ridge Regressor + Placement Tier Pipeline

Model Classification Accuracy: 94.20% | R2 Score: 0.9853 | MAE: 0.98%

1. Dataset Information & Overview

The project uses a benchmark reference dataset comprising 2,500 student profile records across 22 columns. Evaluated features include CGPA, LeetCode rating, solved count, GitHub repositories, commits, stars, LinkedIn connections, certifications, post activity frequency, real-world project count, and project complexity rating.

Reference Dataset Export Files:

- placement_dataset_info.xlsx (Excel Document with sample records, statistical summaries, and tier breakdowns)
- placement_dataset.csv (Raw Dataset CSV File)

2. Machine Learning Algorithms Evaluated & Purpose

Algorithm Name	Accuracy / R2	Purpose & Role in Project	Status
Polynomial Ridge Regressor	94.20% (R2=0.98)	Calculates score & maps to readiness tiers	SELECTED
Multi-Class Softmax Classifier	92.00% (F1=0.74)	Direct multinomial log-odds classification	Evaluated
Gradient Boosting Ensemble	93.00% (R2=0.98)	Fits decision tree residual error vectors	Evaluated
K-Nearest Neighbors (KNN)	81.20% (F1=0.66)	Measures Euclidean distance to 7 nearest peers	Evaluated

3. Feature Importance Breakdown

Rank	Feature Attribute	Importance	Predictive Contribution
#1	coding_score	0.2850	LeetCode rating & problem-solving metric
#2	cgpa_score	0.2450	Academic cumulative grade point average
#3	project_score	0.1780	Real-world project count & complexity level
#4	github_score	0.1150	Open-source commits, stars & public repositories
#5	linkedin_score	0.0820	Professional network connections & certifications
#6	skill_score	0.0540	Broad domain tech stack depth
#7	dept_boost	0.0410	Department specialization branch weighting

4. Automated Diagnostics Engine & Feedback Strategy

The application includes an automated feedback rule engine (feedback_engine.py) that compares candidate metrics against corporate placement cutoffs to generate customized Key Strengths (e.g. top 10% LeetCode rating, high CGPA), Key Weaknesses (e.g. low GitHub activity, missing full-stack projects), and an Actionable Improvement Roadmap.